Agile Infrastructure: an updated overview of IaaS at CERN

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Agenda

- Introduction
- Goals & Use Cases
- CERN Contribution
- IaaS Overview
  - OpenStack deployment
  - Work-in-progress
- Current Status & Timeline
- Outlook

Agile Infrastructure - IaaS

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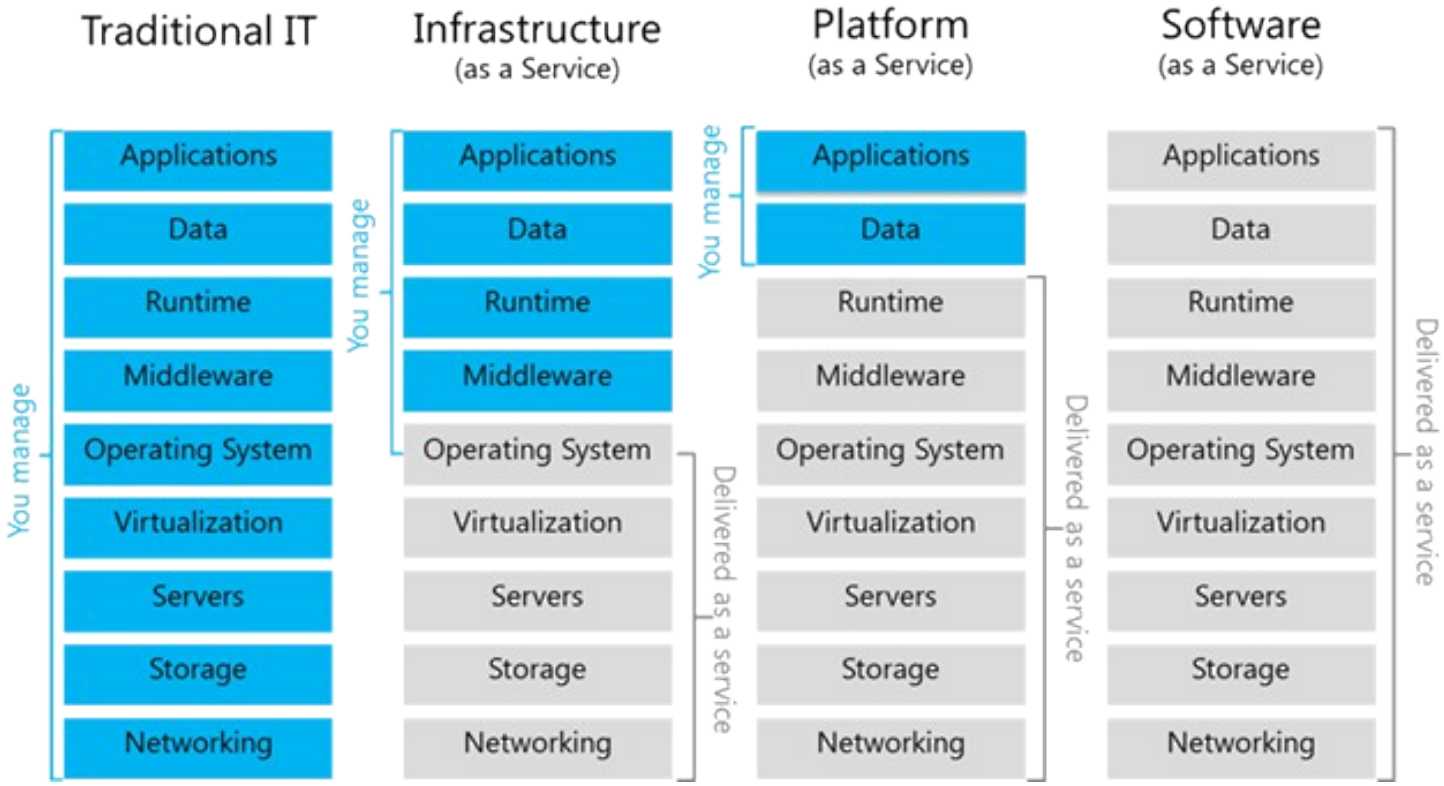
- Rationale
  - Need to manage twice the servers as today
  - No increase in staff numbers
  - …we have to manage the infrastructure efficiently
- Infrastructure as a Service (IaaS)



What is OpenStack?

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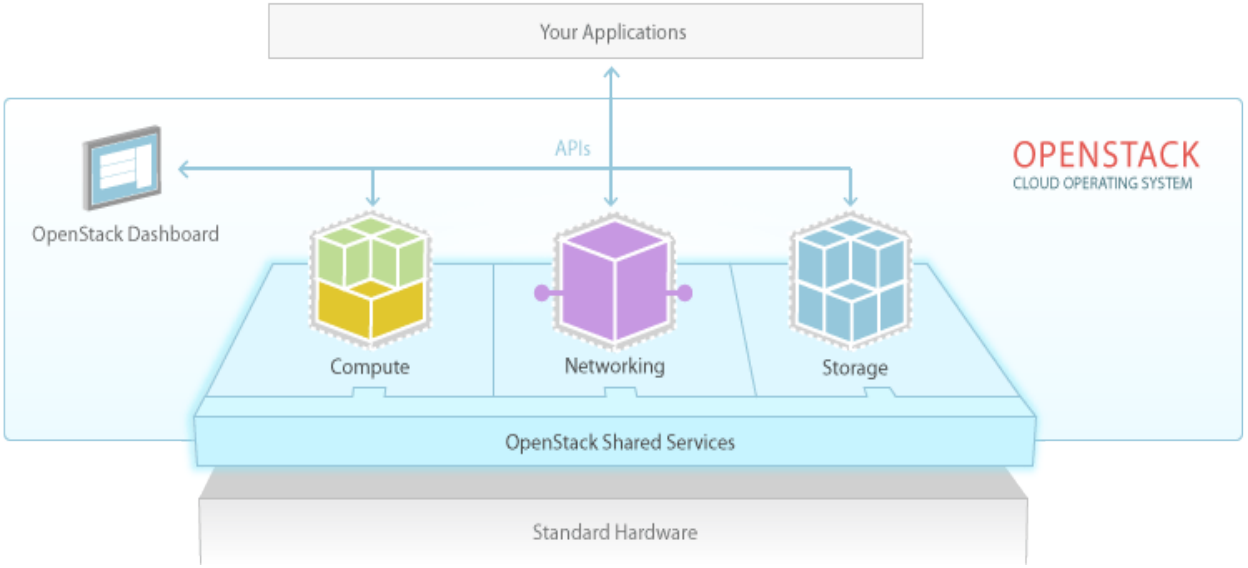
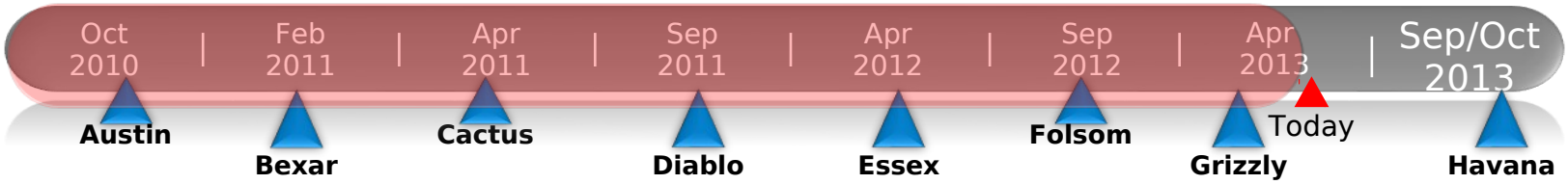
"OpenStack is a cloud operating system that controls large pools of compute, storage, and networking resources throughout a datacenter, all managed through a dashboard that gives administrators control while empowering their users to provision resources through a web interface"



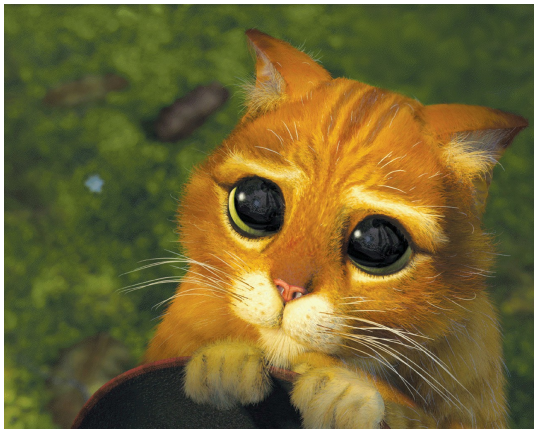
Swift
Nova
Quantum
Keystone
Cinder
Glance
Horizon



Goals

- Objectives
  - 90% of hardware virtualized
  - ~15k hypervisors needed over next two years
- Following industry reference structure
  - Infrastructure, Platform and Applications
- Deploy multi-site
  - Extend to 2nd data center in Hungary
  - Enable remote management
- More efficient use of our hardware
- Better tracking of usage
- Consolidate support to a single device
- Support potential new use cases (PaaS)

Use cases



Pets are given names like pussinboots.cern.ch

They are unique, lovingly hand raised and cared for

When they get ill, you nurse them back to health

Cattle are given numbers like vm0042.cern.ch

They are almost identical to other cattle

When they get ill, you get another one

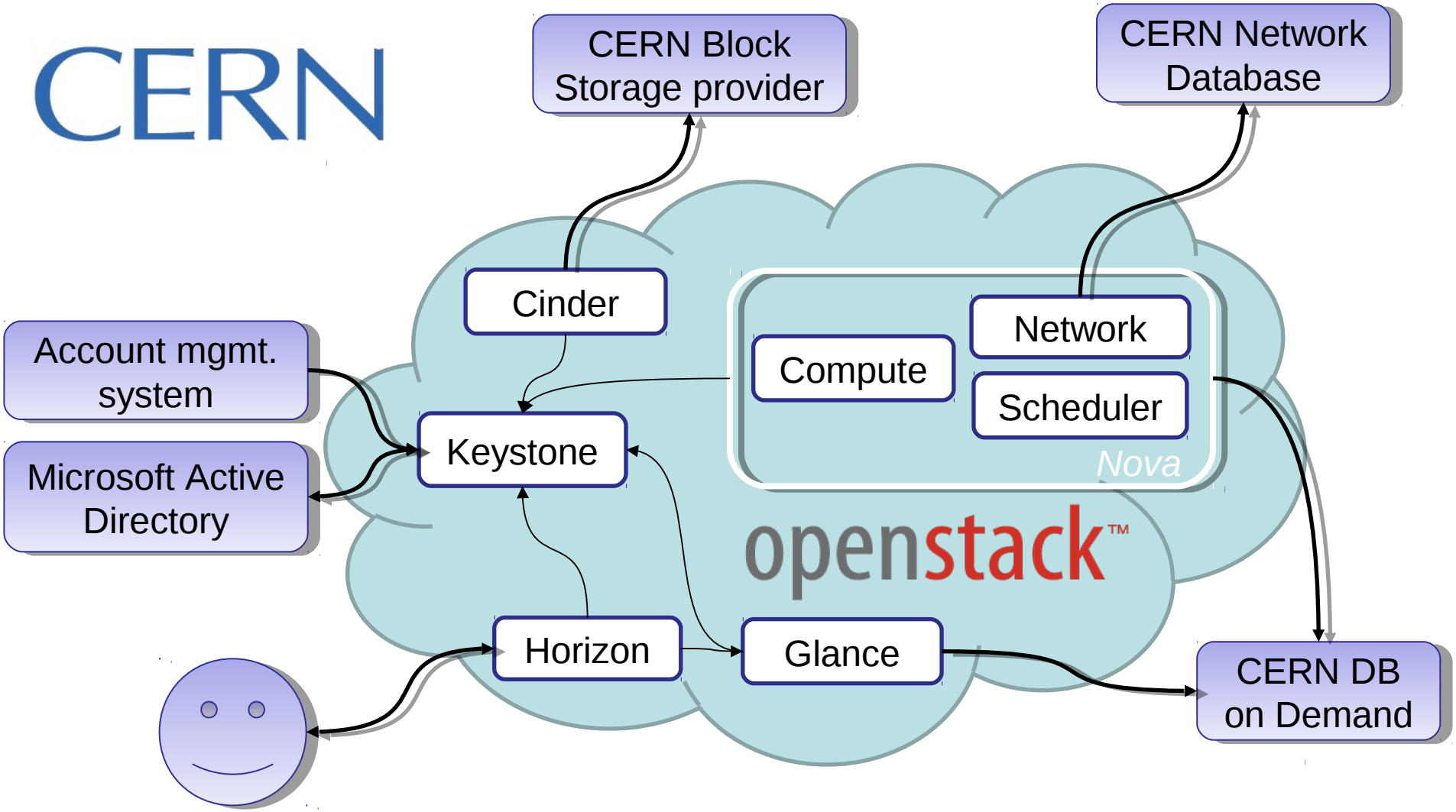
Future application architectures tend towards Cattle but Pet support is needed for some specific zones of the cloud

Collaborating

- CERN is contributing, *the Open Source way*:
  - OpenStack, candidates testing, bug fixing,…
    - Keystone, nova,…
    - Hyper-V Team
  - Fedora, active participation during OpenStack Test Day,…
  - PuppetLabs, Foreman, Aeolus Oz,...

- Exploit the collaboration tools benefits:
  - Mailing lists, IRC, bugzilla, Launchpad, GitHub,…
- Collaborations already starting around
  - BNL, IN2P3, NECTaR (Australia), IHEP (China), ATLAS/CMS Trigger teams,…

Infrastructure Overview



Nova

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Cloud computing fabric controller

- Network manager modified for CERN
  - Integration with network database
  - These problems are specific to our case
  - No modifications pushed upstream
- Nova Compute aware of CERN DNS and AD service
- Multiple availability zones
  - Special zone for Hyper-V hypervisors
  - Scheduler has an additional filter based on image distribution metadata
- Other small tweaks…
  - Renaming while launching multiple instances…

Glance

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Services for discovering, registering, and retrieving vm images

- No major modifications in this module
  - New image meta-data for CERN integration
- Aim for automated image creation and updating
  - Design common process for Linux & Windows images.
  - Use of common tools, basically Aeolus Oz
    - Contribution for improving Windows support in Oz
  - CERN tool for communicate Oz and Glance API
- AI will provide images for CERN supported OS:
  - SLC5, SLC6, Windows {2008 R2, 7, 2012}
  - Users can upload their own images
- Deployed images are contextualized with cloud-init
  - New cloud-init for Windows implemented by Cloudbase

Keystone

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Identity service: authentication, authorization and service catalog

- Full integration with Active Directory via LDAP
  - CERN's AD: 44,000 users & 29,000 groups
  - Uses the LDAP backend with some particular configuration settings
  - Aim for minimal changes to Active Directory
  - Patches submitted around hard coded values, additional filtering and pagination
- Account Mgmt. System integration for project creation/deletion
  - Implemented generic connector to interact with keystone
- Connections are made through secure channels (SSL)
- Now in use in our pre-production instance
  - Map project roles (admins, members) to groups
  - Documentation in the OpenStack.Org wiki

Horizon (I)

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Provides a graphical user interface for managing OpenStack services

- Secure connections through SSL
- Working on integration with CERN SSO system
  - Shibboleth authentication
- Customization for end-users:
  - CERN specific
  - Make it easier for the end user: select OS, flavor, launch!
  - Hide details of security groups, floating Ips, snapshots, etc.
  - Make use of Horizon modularity.

Horizon (II)

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Overview

Logged in as: lfernand Settings Help Sign Out

Project Admin

System Panel

- Overview
- Instances
- Volumes
- Services
- Flavors
- Images
- Projects
- Users

Select a month to query its usage:

April 2013 Submit

Active Instances: 365 **Active RAM:** 2TB **This Month's VCPU-Hours:** 70775.81 **This Month's GB-Hours:** 3978503.25

Usage Summary

Download CSV Summary

Project Name	VCPUs	Disk	RAM	VCPU Hours	Disk GB Hours
ATLAS Victoria	408	9180	816GB	5189.63	896002.59
ATLAS	56	1120	112GB	2794.98	223598.42
IT SVN	52	1040	104GB	1445.59	115647.37
IT Batch	49	980	98GB	886.13	28934.22
Personal straylen	48	960	96GB	5973.34	238933.69
PH-SFT build	44	660	88GB	2455.78	147346.55
NA61	43	860	86GB	1562.77	191997.01
CMS	41	960	82GB	2796.92	193565.16

In progress… Ceilometer

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Unique point of contact to acquire all counters across all OpenStack components

- CERN delivers resources in form of virtual machines and via traditional batch and Grid computing
  - Accounting should cover both use cases
- Interesting metrics for us: *What is the resource usage of experiment A during December? What is the resource usage of user B last year?*

- Use of ceilometer as single source of truth for accounting data
- Batch data is put in the ceilometer database for accounting purpose

- The grow rate of the mongodb database is about 2GB/day
- Hyper-V accounting is expected during Havana release

In progress… Monitoring

- Infrastructure monitoring
  - Cloud hardware and services
  - User's cloud ecosystem
  - Performance of cloud resources
- No OpenStack project
  - Service probing, operator dashboard,…
- Integration with General Notification Infrastructure (GNI) developed by AI Monitoring group
- Windows compute nodes
  - Integrate windows nova nodes in CERN windows monitoring: Microsoft SCOM
  - Development of SCOM Management Pack for OpenStack Nova Compute.
  - Deliver SCOM messages to GNI

Windows deployment

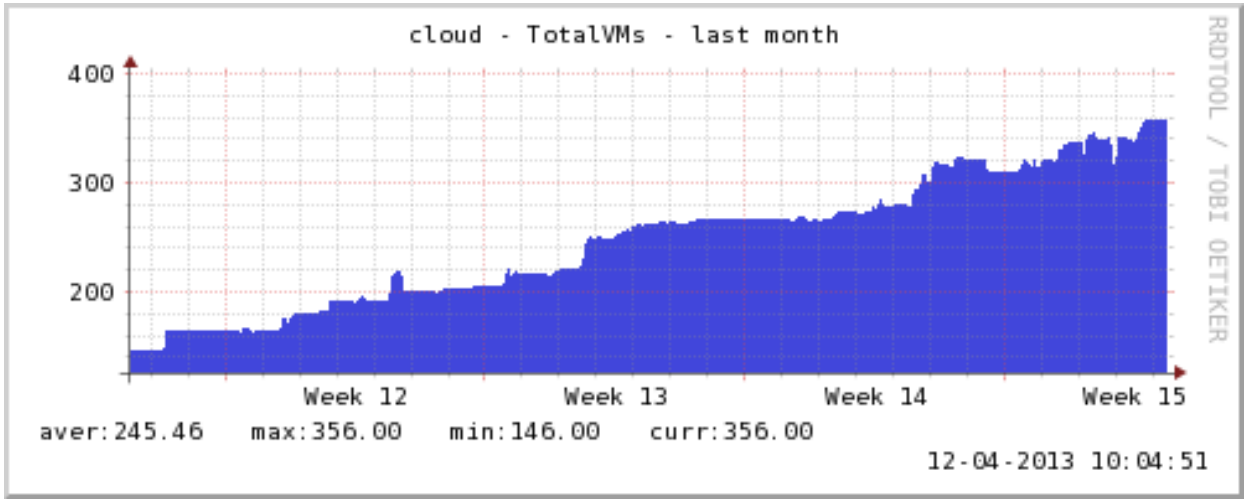
- Deployment of Hyper-V Nova Compute nodes
- Why emphasize Hyper-V?
  - Migration of current virtualization infrastructure based on SCVMM to OpenStack. ~1000 Windows instances
  - Deployment of new Windows machines in a fully Microsoft supported stack
- Aim for deploying Windows compute nodes with Puppet modules
- Working closely with the Hyper-V OpenStack team
  - Puppet modules, cloud-init, rdp console design,...
- In progress: Monitoring, Ceilometer agent...

Service Implementation

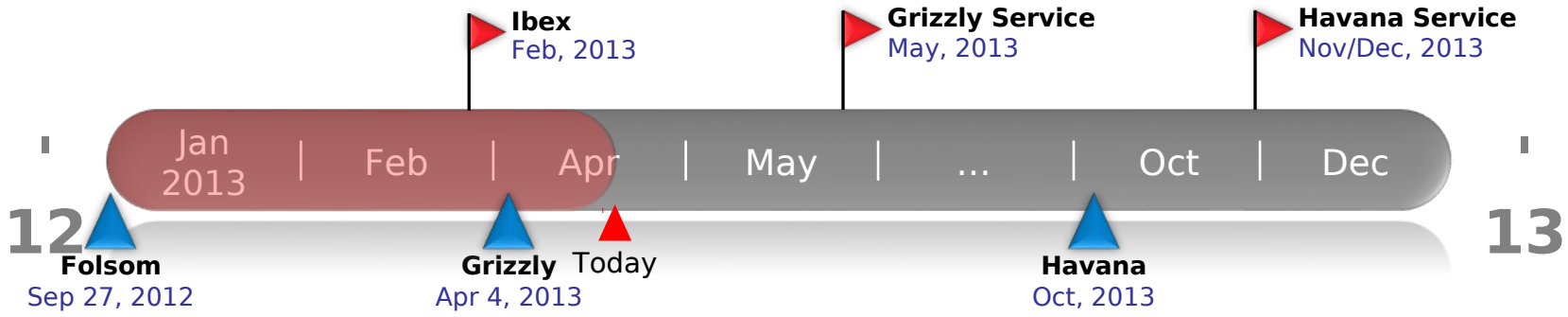
- Preparation for production
  - First steps towards high availability deployment
- OpenStack services running behind DNS load balancer:
  - Two servers
  - Keystone, Network, Scheduler…
  - Working on glance and console
- Database provided by CERN Database Services
  - Database on Demand
  - MySQL
  - External database for: keystone, nova and glance

Current status

- Working on Folsom code base from the EPEL repository
  - Excellent experience with the Fedora Cloud SIG team
- Pre-production facility with around 200 Hypervisors
- Number of VMs growing every day!
  - ~360 instances
  - 164 projects (personal & shared)



Timeline



- Maintain the project linked to upstream releases
- Deployment delay:
  - Time for testing
  - Checking new features suitability for us
  - CERN patches
- We have been delivering a series of pre-production services…
  - Adding functionality…
  - Scaling out…
  - Integrating into IT infrastructure…

Outlook (I)

- Going to production for managed servers
  - End of May, 2013
  - Based on Grizzly release
  - ~2000 Hypervisors (KVM and Hyper-V)
  - Load Balancing
- Grizzly new features
  - Preview release of cells functionality
  - No-DB-Compute
  - Keystone Domains
  - Bare metal provisioning
  - Better Hyper-V support, resizing, cinder, quantum…
  - Database archiving, admin APIs, UI enhancements,…

Outlook (II)

- Non-managed machines, missing requirements:
  - Shared storage for volume service and live-migration
- Investigating alternatives for block storage:
  - NetApp
  - Ceph
- Special attention to cells
  - Exploit cells functionality for scalability
  - Multi-site operations with new data centre
- And more:
  - Load balancing as a service
  - Heat for Orchestration
  - …

Thanks! Questions?

More information…

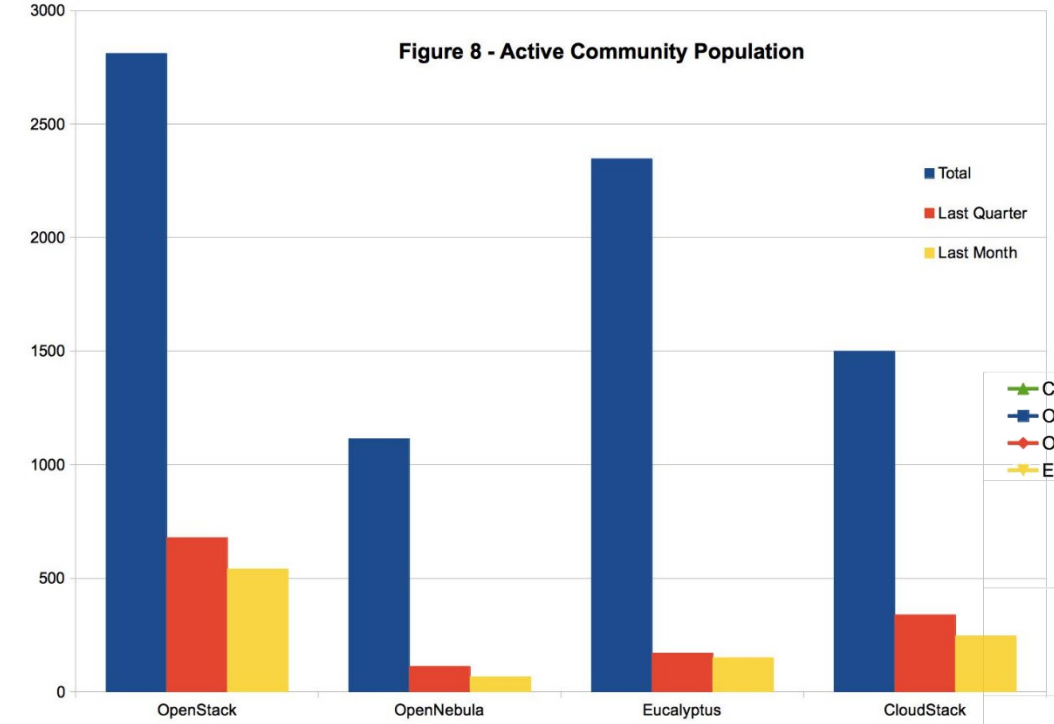
http://cern.ch/openstack

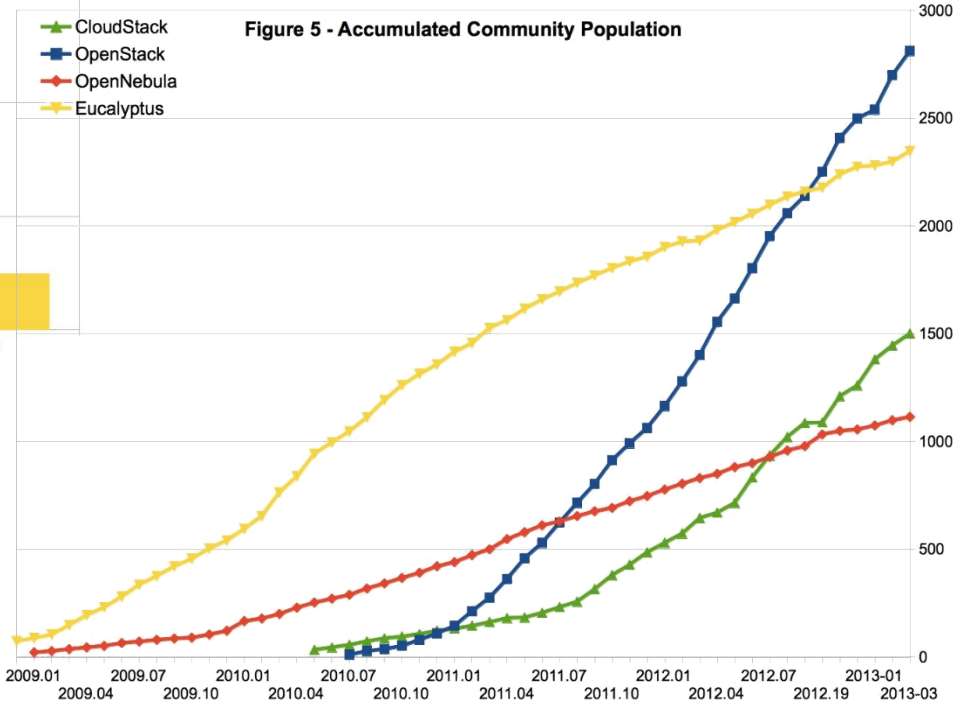
…@ HEPiX Spring 2013…

- *Agile Infrastructure Monitoring*. Pedro Manuel RODRIGUES DE SOUSA ANDRADE
- *High Availability Load Balancing in the Agile Infrastructure (CERN)*. Vaggelis ATLIDAKIS
- *Experiences running a production Puppet infrastructure at CERN*. Ben JONES
- *Ceph: A scalable, organic option for Storage-as-a-Service at CERN*. Arne WIEBALCK

Backup Slides

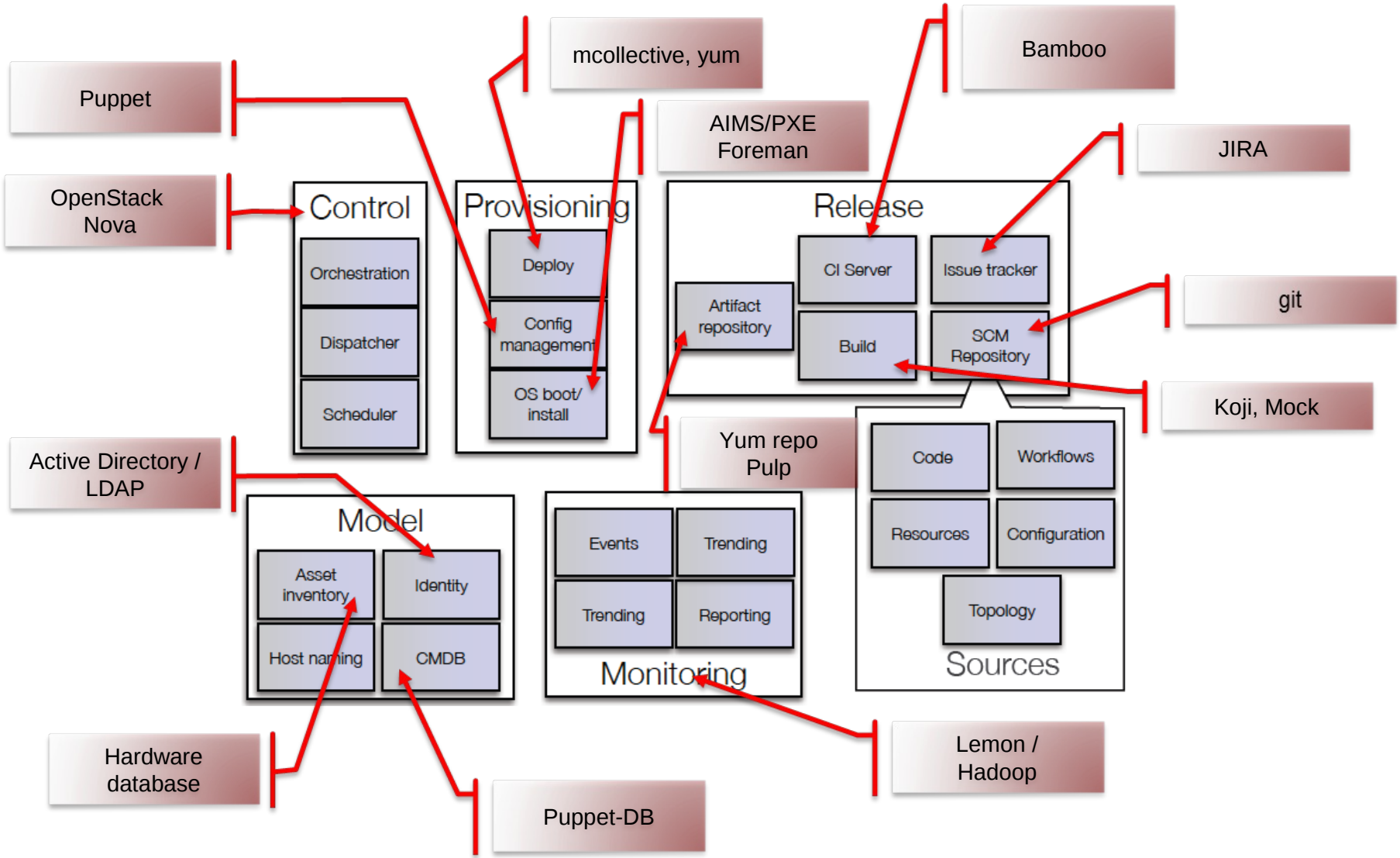
IaaS Community Analysis

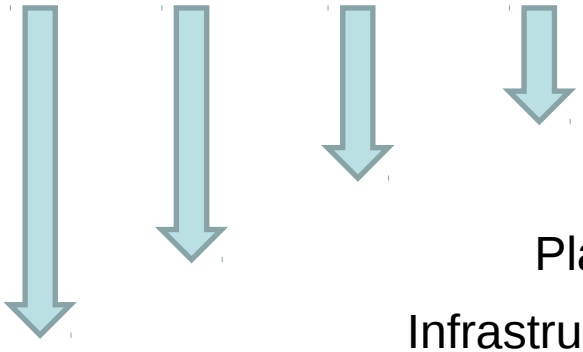




http://www.qyjohn.net/?p=3120

Building Blocks



IT as a Cloud Service Provider ?

- Following industry reference structure
  - Infrastructure, Platform and Applications
- Start first with Infrastructure-as-a-Service at scale
  - Just provide Amazon functionality
- Value add increases up the chain so we need a vibrant ecosystem for PaaS and SaaS